

Analysis PhD Exam, January 2024

Write solutions in a neat and logical fashion, giving complete reasons for all steps. Attempt SIX problems.

1. Let $(X, \mathcal{M}, \mu)$ be a finite measure space. Let (f_n) be a sequence of measurable functions and suppose that $\sum_n \int |f_n| d\mu < \infty$. Consider the series $\sum_n f_n(x)$. Does this series converge... Prove your claims.
 - a) pointwise a.e.?
 - b) in measure?
 - c) in the L^1 norm?
2. State Tonelli's theorem, and give an example to show that the σ -finiteness hypothesis is necessary.
3. This problem has two parts.
 - a) Let ν be a signed measure on $(X, \mathcal{M})$. Define what it means for a set to be positive, negative or null for ν , and state the Hahn decomposition theorem.
 - b) Prove that if ν is a signed measure, then there exist unique positive measures ν^+, ν^- so that $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$.
4. Let $\varphi_n : \mathbb{R} \rightarrow \mathbb{C}$ be a sequence of Lebesgue measurable functions and suppose that there is a number M such that $\|\varphi_n\|_\infty \leq M$ for all n . Prove that if

$$\int_a^b \varphi_n(t) dt \rightarrow 0$$

for every interval $[a, b] \subset \mathbb{R}$, then

$$\int_{-\infty}^{\infty} \varphi_n(t) f(t) dt \rightarrow 0$$

for every $f \in L^1(\mathbb{R})$.

5. Let μ and ν be finite, positive measures defined on the same measurable space $(X, \mathcal{M})$, and suppose that $\nu \ll \mu$.
 - a) Prove that if $L^1(\mu) \subset L^1(\nu)$, then the inclusion map $\iota : L^1(\mu) \hookrightarrow L^1(\nu)$ is necessarily bounded.
 - b) Prove that if $L^1(\mu) \subset L^1(\nu)$ then $\frac{d\nu}{d\mu} \in L^\infty(\mu)$.
6. Let $\mathcal{X}, \mathcal{Y}$ be Banach spaces and $T : \mathcal{X} \rightarrow \mathcal{Y}$ a linear map. Prove that if $f \circ T \in \mathcal{X}^*$ for all $f \in \mathcal{Y}^*$, then T is bounded.
7. Short answer.
 - a) Explain why there is no norm on c_{00} (the space of finitely nonzero sequences) which makes it into a Banach space.
 - b) Explain why there is no bounded, surjective linear map from c_0 onto ℓ^∞ .
 - c) Explain why the norm in $L^1(\mathbb{R})$ cannot be given by an inner product.
8. Let H be a Hilbert space with orthonormal basis $\{e_k\}_{k=1}^\infty$. Let (h_n) be a sequence in H and let $h \in H$. Prove that the following statements are equivalent:

- 1) $\langle h_n, g \rangle \rightarrow \langle h, g \rangle$ for all $g \in H$.
- 2) $\langle h_n, e_k \rangle \rightarrow \langle h, e_k \rangle$ for all $k = 1, 2, \dots$, AND $\sup_n \|h_n\| < \infty$.